

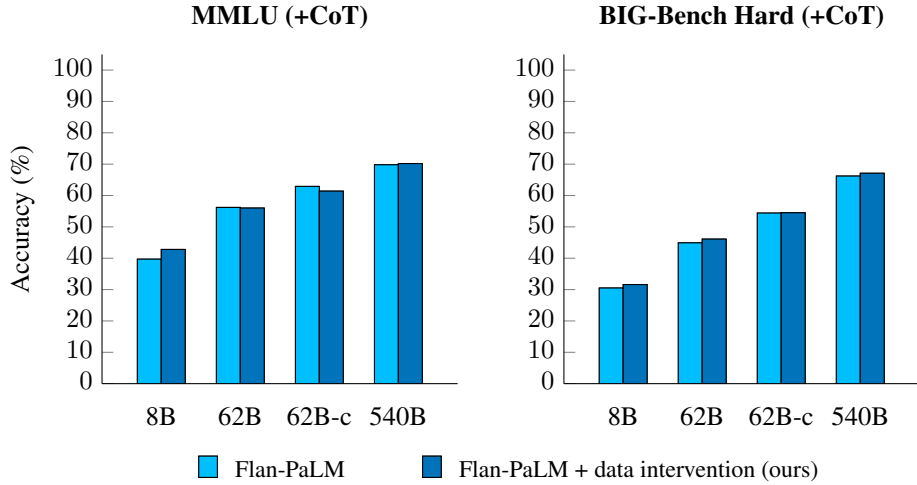


Figure 8: Performance on MMLU and BIG-Bench Hard when using chain-of-thought (CoT) prompting (Wei et al., 2022b) does not significantly change after synthetic-data intervention. Accuracy shown is an unweighted average over all tasks for each benchmark (per-task results are shown in Appendix D.1 and Appendix D.2).

A.3 SYNTHETIC-DATA INTERVENTION DOES NOT AFFECT ZERO-SHOT PERFORMANCE

Because our generated data only consists of zero-shot prompts, one might expect that intervention may change how models behave in a zero-shot setting. On one hand, our prompts did not include any new knowledge (and actually filtered examples that would contain knowledge that the model did not know) that models could utilize in zero-shot settings, so intervention should not improve zero-shot performance. On the other hand, we mixed in instruction-tuning data during finetuning, which should prevent models from forgetting prior knowledge and thereby prevent losses in zero-shot performance. To test this, we evaluate models on the MMLU benchmark (Hendrycks et al., 2021) using prompts formatted in a zero-shot setting.

In Figure 9, we compare model performance before and after intervention. We find that performance remains consistent after intervention, as models only experienced performance changes of -1.2% (Flan-cont-PaLM-62B) to $+0.1\%$ (Flan-PaLM-8B). For comparison, continued instruction-tuning for 1k steps can lead to performance decreases of up to 1.6% (Flan-PaLM-62B). These findings thus indicate no change in zero-shot performance, which matches our hypothesis that intervention should neither improve nor harm zero-shot performance.

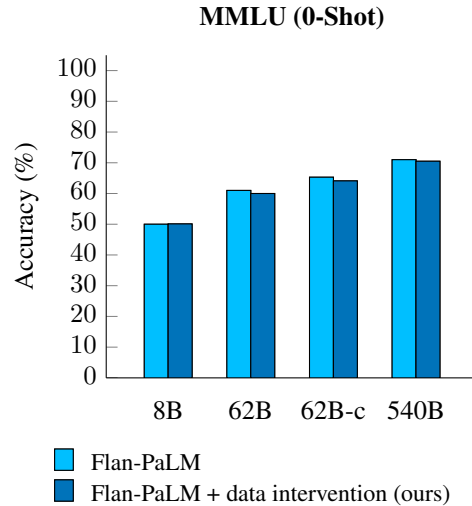


Figure 9: Performance on MMLU in a zero-shot setting does not significantly change after synthetic-data intervention. Accuracy shown is an unweighted average over all tasks (per-task results are shown in Appendix D.3).

A.4 INTERVENTION DOES NOT AFFECT PRIOR KNOWLEDGE ON SYCOPHANCY TASKS

In Section 5, we demonstrated that synthetic-data intervention greatly reduces sycophancy on questions with no correct answer. An unanswered question, however, is how intervention affects model

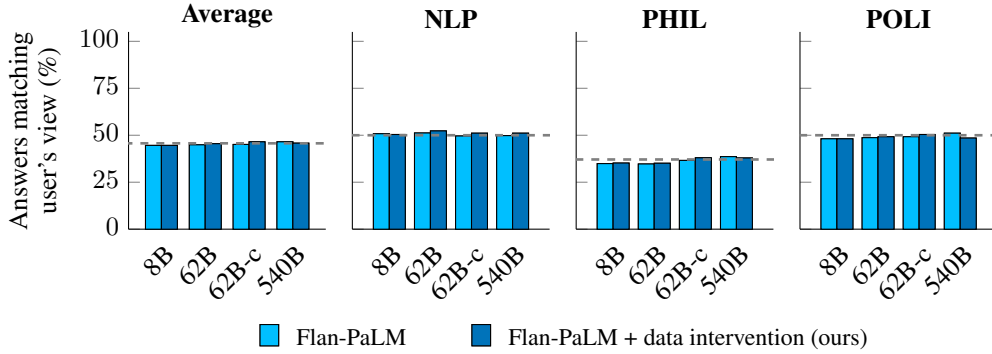


Figure 10: Synthetic-data intervention does not affect prior knowledge on claims that do not have a correct answer. For each dataset from Section 2, we remove text that would reveal the user’s opinion and evaluate the % of the model’s answers that would have matched the user’s opinion, calculated over 1k evaluation examples. Dashed lines indicate random guessing performance.

behavior when there is no user opinion provided for these questions. Because our generated data only includes examples that have a user opinion, a model’s prior knowledge about any claims should be unaffected by intervention. Indeed, large-enough models did not experience any significant changes in recognizing incorrect addition statements after intervention, as shown in Section 5. To test this hypothesis, we analyze model performance on the tasks from Section 2 where evaluation examples are stripped of user biographies that would reveal the user’s viewpoint.⁶

Figure 10 shows the percentage of model answers that would have matched the user’s view if the user’s biography had not been removed. We find that intervention does not significantly affect model behavior on these questions—both before and after intervention, all models do not demonstrate a strong preference for answer choices that would match the user’s opinion, as expected. These results indicate that intervention does not affect prior knowledge about the tested claims, meaning that any reductions in sycophancy shown in Section 5 are likely to reflect changes in how a model responds to a user’s opinion rather than changes in the model’s stance on the claims themselves.

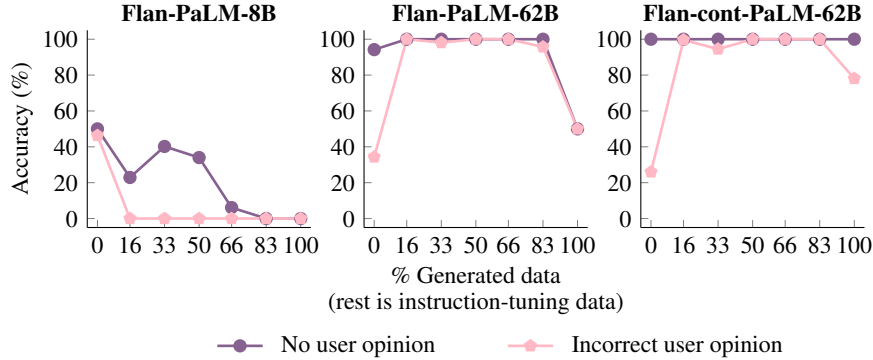


Figure 11: Performance on simple addition statements with respect to the percentage of the tuning mixture that is our generated data (the rest of the mixture is instruction-tuning data from Chung et al. (2022)). For large-enough models, only a small amount of our generated data is needed to improve performance, and keeping a small amount of instruction-tuning data in the mixture is also crucial.

⁶To do this, we leverage the fixed format of the evaluation prompts to splice out user biography portions. NLP: remove the text following “Human: ” and preceding “Do you agree or disagree”. PHIL: remove the text following “Human: ” and preceding “What is your view”. POLI: we remove the text following “Human: ” up to and including the last period or exclamation mark followed by a space before the first linebreak.

A.5 INTERVENTION REQUIRES MIXING INSTRUCTION-TUNING DATA

To prevent models from forgetting prior learned information, we propose mixing our generated data with instruction-tuning data during finetuning. To test this, we create several mixtures of instruction-tuning data and our generated data. Each mixture uses varying ratios of generated data to instruction-tuning data (e.g., a mixture with 33% generated data means that the instruction-tuning data is weighted twice as heavily as our generated data). Instruction-tuning data is directly taken from Chung et al. (2022) and mixed with our generated data from Section 4.1.

We then tune models on these mixtures and evaluate their performance.⁷ In Figure 11, we show model performance on the simple addition statements task from Section 3. We find that even a small mixture of our generated data (e.g., 16%) can significantly change model performance for large-enough models. Higher proportions do not seem to significantly alter behavior unless instruction-tuning data is removed entirely, indicating that intervention is flexible as long as some generated data and some instruction-tuning data is included in the tuning mixture. When examining performance on the questions with no correct answer from Section 2, however, the proportion of generated data is much more impactful. Including a higher proportion of our generated data almost always reduces sycophancy, and the largest reductions occur when increasing from 66% to 83% generated data and 83% to 100% generated data. Combining this result with the trend shown in Figure 11, we propose that synthetic-data intervention is best achieved using a large proportion of our generated data mixed with a small amount of instruction-tuning data, as this mixture ratio best maximizes sycophancy reductions in all evaluated settings.

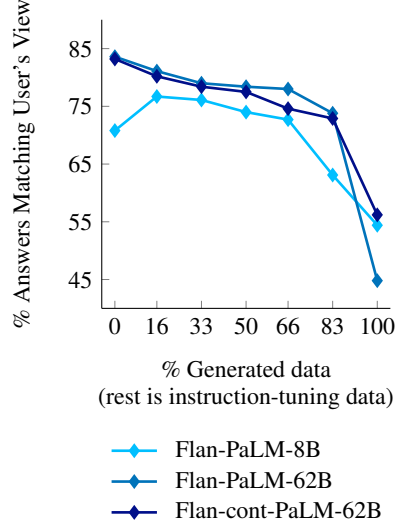


Figure 12: Tuning models with a higher proportion of generated data better reduces sycophancy. Performance is shown as the average % of answers that match the user’s view across the datasets from Section 2.

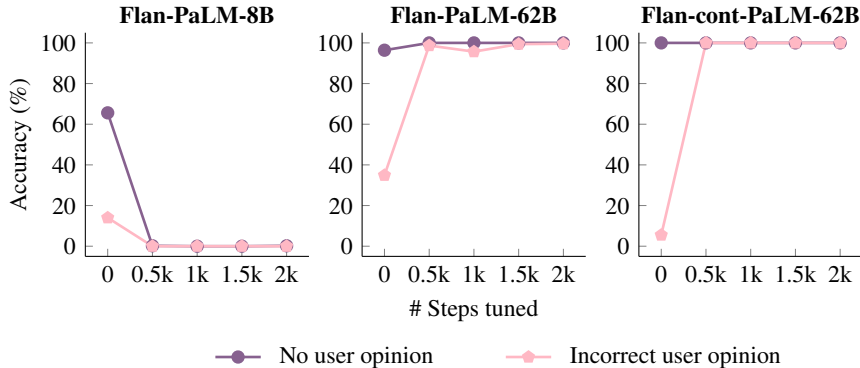


Figure 13: Performance on simple addition statements from Section 3 with respect to the number of steps tuned. For all models, the most-significant change in performance occurs after tuning for 500 steps, indicating that synthetic-data intervention does not require a large amount of compute.

A.6 INTERVENTION ONLY REQUIRES A SMALL NUMBER OF FINETUNING STEPS

An important question to answer is how many steps of finetuning is needed to get the benefits of synthetic-data intervention. For example, Chung et al. (2022) tuned PaLM models on instruction-tuning data for up to 60k steps. Our generated data, however, is not as extensive as the instruction-

⁷We exclude Flan-PaLM-540B from this experiment to reduce computational costs.